Exp No: 13

EMBEDDED C PROGRAM TO ADD AN ARRAY OF 16-BIT NUMBERS

AND STORE THE 32-BIT RESULT IN INTERNAL RAM

AIM:

To write an Embedded C program to add an array of 16-bit numbers and store the results in

internal RAM using Keil software.

SOFTWARE REQUIRED:

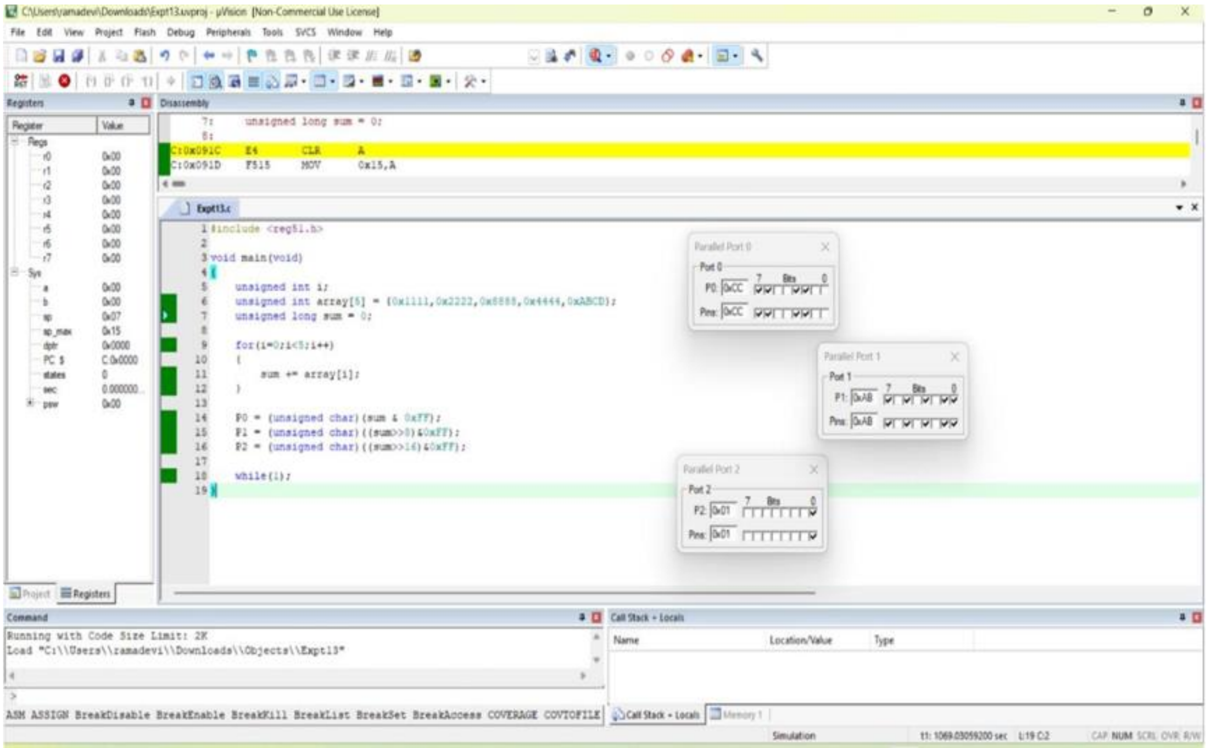
• Keil

PROGRAM:

```
#include &lt;reg51.h&gt;

void main(void)

{

unsigned int i;

array[00x00 unsigned int array[5] = {0x1111, 0x2222, 0x8888, 0x4444, 0xABCD};

unsigned long sum = 0;

for (i = 0; i &lt; 5; i++)

{

sum = sum + array[i];

}

P0 = (unsigned char)(sum &amp; 0xFF); // Lower 8 bits

P1 = (unsigned char)((sum &gt;&gt; 8) &amp; 0xFF); // Next 8 bits

P2 = (unsigned char)((sum &gt;&gt; 16) &amp; 0xFF); // Upper 8 bits
```

```
while(1); // keep output stable

}
```

OUTPUT:



RESULT:

Thus the program has been successfully verified and executed.